04/11/2026

Code-B

Aakash
Medical | IIT-JEE | Foundations

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MM : 720

Fortnightly Test for NEET-2027 (XII Passed) Ph-2 - FT06B

Time : 180 Min.

Topics Covered:
Physics : Oscillations, Waves, Electric Charges & Fields
Chemistry : Redox Reactions, Electrochemistry, Solutions
Botany : Plant Kingdom
Zoology : Chemical Coordination & Integration

General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

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PHYSICS

1. A particle performs SHM with time period T. Kinetic energy ($K(t)$) and potential energy ($U(t)$) of particle are such that
(1) Both $K(t)$ and $U(t)$ have same time period T
(2) Both $K(t)$ and $U(t)$ have same time period $\frac{T}{2}$
(3) Only $K(t)$ has time period T
(4) Only $U(t)$ has time period T

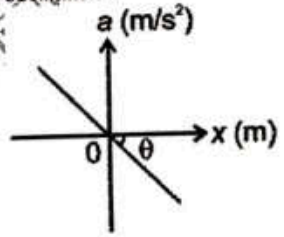
2. The phase difference (in radian) between displacement and velocity in SHM is
(1) π (2) 2π
(3) $\frac{\pi}{2}$ (4) $\frac{\pi}{4}$

3. A uniform string is hanging from a rigid support. A transverse pulse is generated at its free end. The speed at which the pulse travels at a distance x from lower end is proportional to
(1) $\sqrt{x}$ (2) $\frac{1}{x}$
(3) $\frac{1}{\sqrt{x}}$ (4) $x^{3/2}$

4. Electric field of an infinitely long uniform charged straight wire having linear charge density λ is proportional to (r is perpendicular distance from the wire)
(1) r (2) $\frac{1}{r^2}$
(3) $\frac{1}{r^3}$ (4) $\frac{1}{r}$

5. In the uniform electric field, an electric dipole is placed perpendicular to electric field. Then
(1) The dipole is in unstable equilibrium position
(2) The potential energy of dipole is minimum
(3) The potential energy of dipole is zero
(4) (2) and (3) are correct

6. Acceleration-displacement (a-x) graph of a particle executing S.H.M. is shown in the figure. If $\tan\theta = 8$, the frequency of oscillation is



(1) $\frac{2}{\pi}$ Hz (2) $\frac{\sqrt{2}}{\pi}$ Hz
(3) $\frac{2\sqrt{2}}{\pi}$ Hz (4) $\frac{1}{\pi}$ Hz

7. A neutral body is charged positively by rubbing it, its weight
(1) Increases slightly
(2) Decreases slightly
(3) Remains constant
(4) May increase or decrease

8. In a stationary wave produced in a string, all the particles
(1) On either side of a node vibrate in the same phase.
(2) In the region between two consecutive nodes vibrates in the same phase.
(3) In the region between two antinodes vibrate in the same phase.
(4) Of the medium vibrate in the same phase.

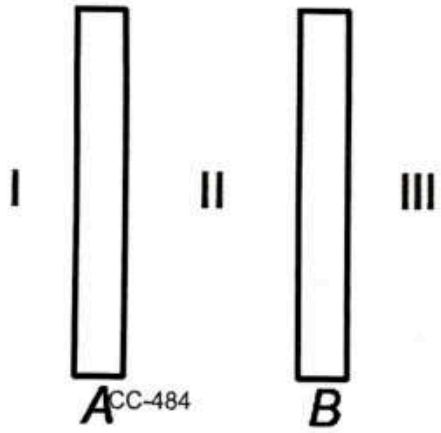
9. Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**.
Assertion A: A combination of two simple harmonic motion with arbitrary amplitudes and phases is necessarily simple harmonic motion.
Reason R: A combination of two simple harmonic motions is periodic only if frequency of one motion is an integral multiple of other's frequency.
In the light of the above statements, choose the correct answer from the options given below.
(1) Both **A** and **R** are true and **R** is the correct explanation of **A**
(2) Both **A** and **R** are true but **R** is not the correct explanation of **A**
(3) **A** is true but **R** is false
(4) Both **A** and **R** are false

10. The electric field at a distance $\frac{5R}{2}$ from the centre of a charged conducting spherical shell of radius R is E_0. The electric field at a distance $\frac{3R}{4}$ from the centre of the shell is
(1) Zero (2) E_0
(3) $\frac{9E_0}{5}$ (4) $\frac{3E_0}{5}$

11. The equation for a simple harmonic plane progressive wave propagating along the positive direction of x-axis is (Symbols have their usual meanings)
(1) $y = A\sin(\omega t + kx)$ (2) $y = A\sin k(vt + x)$
(3) $y = A\sin 2\pi\left(\frac{t}{T} - \frac{x}{\lambda}\right)$ (4) $y = A\sin 2\pi\left(\frac{t}{T} + \frac{2x}{\lambda}\right)$

12. In a closed organ pipe the harmonic associated with p^{th} overtone will be
(1) $2p + 1$ (2) $2p - 1$
(3) $p + 1$ (4) $p - 1$

13. The given figure shows, two parallel plates A and B of charge densities $+3\sigma$ and -3σ respectively. Electric intensity will be zero in the



(1) Region I and II (2) Region II and III
(3) Region II (4) Region I and III

14. A short dipole is at origin and directed towards positive x-axis. The electric field at a point $A(0, a, 0)$ is $\vec{E}$. The electric field at a point $B(a, 0, 0)$ will be
(1) $\vec{E}$ (2) $2\vec{E}$
(3) $-\frac{\vec{E}}{2}$ (4) $-2\vec{E}$

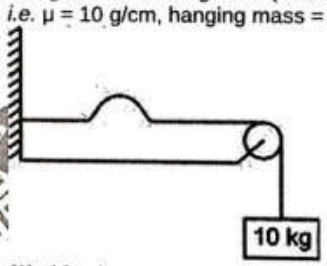
15. Consider the following statements A and B and identify the correct answer.
A: In a transverse wave, energy and matter are transferred from one point to the other.
B: In a stationary wave, the energy does not flow from one side to other side of a node.
(1) A is correct but B is incorrect
(2) A is incorrect but B is correct
(3) Both A and B are correct
(4) Both A and B are incorrect

16. Frequency of second harmonic of a open organ pipe of length 100 cm, if velocity of sound in air is 330 m/s, is
(1) 165 Hz (2) 330 Hz
(3) 660 Hz (4) 500 Hz

17. A particle is under a motion governed by, $y = [4\sin(8t) - 3\cos(8t)]$ (Where y is in cm and t is in seconds). Time period of the simple harmonic motion governed by the particle is
(1) $\frac{\pi}{3}$ s (2) $\frac{\pi}{4}$ s
(3) $\frac{\pi}{2}$ s (4) $\frac{\pi}{5}$ s

18. If displacement of a particle in SHM is given by x, then the total energy (E) of the particle is related as
(1) $E \propto x$ (2) $E \propto x^{1/2}$
(3) $E \propto x^2$ (4) $E \propto x$

19. The speed of the wave pulse generated on the stretched string as shown in figure is (mass per unit length of the string *i.e.* μ = 10 g/cm, hanging mass = 10 kg)



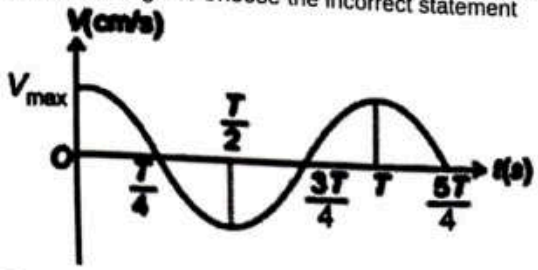
(1) 10 m/s (2) $\sqrt{10}$ m/s
(3) 1 m/s (4) 0.1 m/s

20. Consider the following statements and choose the correct option.
Statement I: Value of charge increases with increase in its speed.
Statement II: Charge on any object is quantised.
Statement III: For an isolated system net charge is conserved.
Statement IV: Specific charge on any object is invariant.
(1) Only statement II is correct
(2) Only statements II and III are correct
(3) Only statements I, II and III are correct
(4) Only statements II, III and IV are correct

21. The magnitude of electric field strength E such that an electron placed in it would experience an electrical force equal to its weight is given by (m = mass of electron, e = charge on electron, g = acceleration due to gravity)
(1) mge (2) $\frac{mg}{e}$
(3) $\frac{e}{mg}$ (4) $\frac{eg}{2m}$

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22. The velocity-time (v-t) graph of a particle executing S.H.M is shown in the figure. Choose the incorrect statement



(1) Force is maximum at $t = \frac{3T}{4}$
(2) Potential energy of the particle is minimum at $t = \frac{T}{4}$
(3) Kinetic energy of the particle is maximum at $t = T$
(4) Acceleration of particle is maximum at $t = \frac{T}{4}$

23. Laplace's correction in the formula for the speed of sound in gases as given by Newton, was needed because sound wave in gases
(1) Propagates adiabatically
(2) Propagates isothermally
(3) Are transverse
(4) Have long wavelength

24. The motion of a bob of simple pendulum is
(1) SHM for small angular displacement
(2) SHM for large angular displacement
(3) SHM when maximum angular displacement is 90°
(4) SHM for all angular displacement

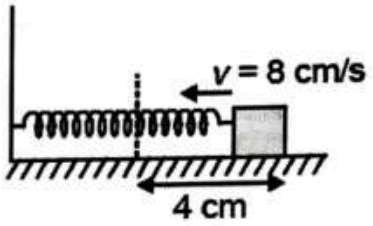
25. First two lengths of air column, in a resonance column method, were found to be 32 cm and 98 cm respectively. It is found that velocity of sound in laboratory is 336 ms^{-1}. The frequency of vibrating tuning fork is
(1) 524 Hz
(2) 416.2 Hz
(3) 316.5 Hz
(4) 254.5 Hz

26. Electrostatic force of interaction between two point charges varies with separation R between them as
(1) $\frac{1}{R^3}$
(2) $\frac{1}{R^2}$
(3) $\frac{1}{R^4}$
(4) $\frac{1}{R}$

27. Consider the following two statements:
Statement I: For small damping, the damped oscillations of a spring-mass system are periodic.
Statement II: In damped SHM, the system loses mechanical energy which is converted into heat in the surrounding medium.
In the light of the above statements, choose the correct option.
(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

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28. A block of mass 0.02 kg is connected with a spring and is free to oscillate on horizontal smooth surface as shown. The angular frequency of oscillation is 2 rad s^{-1}. The block is pulled by 4 cm (from equilibrium position) and then pushed towards the spring with a velocity of 8 cm/s. The amplitude of oscillation is (Neglect any damping)



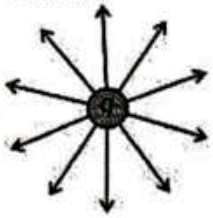
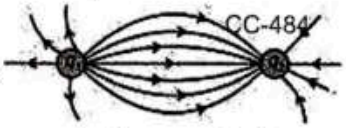
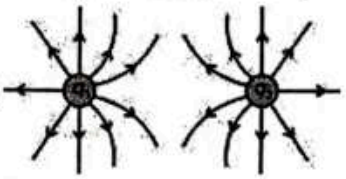
(1) $3\sqrt{2}$ cm
(2) $4\sqrt{2}$ cm
(3) $2\sqrt{2}$ cm
(4) 1 cm

29. Consider a uniform electric field $\vec{E} = 8 \times 10^3 \hat{i}$ N/C. What is the flux through a square of side length 10 cm, if normal to its plane makes an angle 60° with x-axis?
(1) 40 $\frac{Nm^2}{C}$
(2) 80 $\frac{Nm^2}{C}$
(3) 20 $\frac{Nm^2}{C}$
(4) 10 $\frac{Nm^2}{C}$

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30. A wave represented by the equation $y = -a\cos(\omega t + kx)$ is superposed with another wave to form a stationary wave such that point x = 0 is a node. The equation for the other wave is
(1) $-a\sin(kx - \omega t)$
(2) $a\sin(kx - \omega t)$
(3) $a\cos(kx - \omega t)$
(4) $a\cos(\omega t + kx)$

31. Match the correct representation of electric field lines with the correct type of charge mentioned.

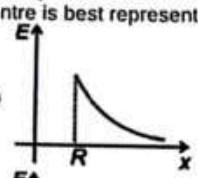
Column-I	Column-II
a. $q < 0$	(i) 
b. $q > 0$	(ii) 
c. $q_1 > 0, q_2 < 0$	(iii) 
d. $q_1 > 0, q_2 > 0$	(iv) 

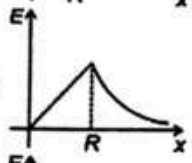
(1) a(i), b(ii), c(iv), d(iii)
(2) a(ii), b(i), c(iv), d(iii)
(3) a(ii), b(i), c(iii), d(iv)
(4) a(i), b(ii), c(iii), d(iv)

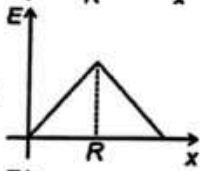
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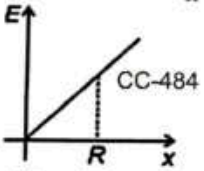
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32. Electric field (E) due to a uniformly charged non-conducting solid sphere of radius R, as a function of distance (x) from its centre is best represented as

(1) 

(2) 

(3) 

(4) 

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33. The differential equation representing the S.H.M. of particle is $16\frac{d^2y}{dt^2}+9y=0$. If particle is at mean position initially, then time taken by the particle to reach half of its amplitude first time will be

(1) $\frac{4\pi}{9}$ s
(2) $\frac{5\pi}{9}$ s
(3) $\frac{\pi}{3}$ s
(4) $\frac{2\pi}{9}$ s

34. Velocity of sound is maximum in

(1) Dry air
(2) Humid air
(3) Water
(4) Iron rod

35. A stationary wave is set-up in the string fixed at both ends, with equation $y=4\sin(8\pi x)\cos(90\pi t)$ (Where y and x are in cm and t in second). Amplitude, wavelength and velocity of the component waves are respectively

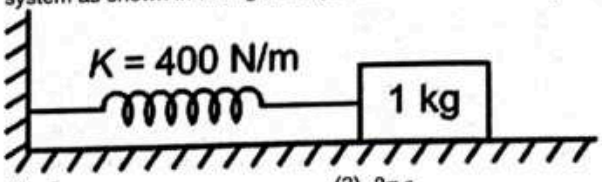
(1) 2 cm, 0.25 cm, 11.25 cms^{-1}
(2) 4 cm, 0.25 cm, 10 cms^{-1}
(3) 3 cm, 25 cm, 10 cms^{-1}
(4) 4 cm, 0.50 cm, 5 cms^{-1}

36. Four charges each of magnitude q are fixed at the four corners of a square having side a. Electric field at the centre of square will be

(1) $\frac{\sqrt{2}}{\pi\varepsilon_0}\frac{q}{a^2}$
(2) $\frac{1}{\pi\varepsilon_0}\frac{q}{\sqrt{3}a^2}$
(3) Zero
(4) $\frac{q}{4\sqrt{3}\pi\varepsilon_0 a^2}$

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37. The time period of small oscillations of the spring mass system as shown in the figure is (all surfaces are smooth)



(1) $\frac{\pi}{10}$ s
(2) 2π s
(3) $\frac{2\pi}{\sqrt{10}}$ s
(4) 10 s

38. For superposition of two harmonic waves one of frequency 12 Hz and the other of frequency 10 Hz, the beat frequency is

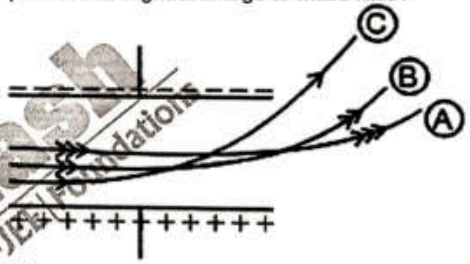
(1) 1 Hz
(2) 2 Hz
(3) 3 Hz
(4) 4 Hz

39. Two parallel metal plates having charges $+Q$ and $-Q$, facing each other at a certain distance are initially dipped in water, then these plates are kept in air keeping charge on the plates and distance between the plates same, then electric field between the plates will

(1) Remain same
(2) Decrease
(3) Becomes zero
(4) Increase

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40. Three particles are projected in a uniform electric field with same velocity perpendicular to the field as shown. Which particle has highest charge to mass ratio?



(1) A
(2) B
(3) C
(4) All have same charge to mass ratio

41. Which of the following statements is true for the velocity and acceleration of a particle executing simple harmonic motion?

(1) When velocity is maximum, acceleration is maximum
(2) When acceleration is zero, velocity can have any value
(3) When velocity is maximum, acceleration is zero
(4) When velocity is minimum, acceleration is minimum

42. A spring mass system has time period T of its oscillations. It is now suspended above a large horizontal metal sheet with uniformly distributed positive charge. If mass is given some positive charge, then its new time period of oscillation will be

(1) T
(2) $> T$
(3) $< T$
(4) Depends on A (amplitude)

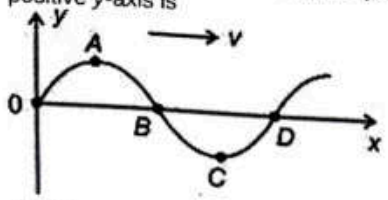
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43. A point charge q is placed at the centre of a cube of side length l. The electric flux through one of the faces of cube is

(1) $\frac{q}{4\varepsilon_0}$
(2) $\frac{q}{6\varepsilon_0}$
(3) $\frac{q}{3\varepsilon_0}$
(4) $\frac{q}{12\varepsilon_0}$

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44. A transverse wave travelling along positive x-direction on a string is shown at an instant. The particle having velocity in positive y-axis is



(1) A
(2) B
(3) C
(4) D

45. Motion of a particle undergoing uniform circular motion is
(1) Periodic and oscillatory
(2) Periodic but non-oscillatory
(3) Non-periodic and non-oscillatory
(4) Non-periodic and oscillatory

CHEMISTRY

46. If oxidation numbers of A, B, C are +1, +4, –3 respectively then possible formula of compound will be
(1) ABC
(2) A_3BC_2
(3) ABC_3
(4) A_2BC_2

47. Consider the following statements about Galvanic cell.
(a) The flow of current is from cathode to anode.
(b) Electrons produced at the cathode and travel through the external circuit to the anode.
(c) Anode is negatively charged electrode while cathode is positively charged electrode.
The incorrect statement(s) is/are
(1) (a) and (c) only
(2) (a), (b) and (c)
(3) (b) only
(4) (b) and (c) only

48. Choose the correct oxidation state(s) of sulphur atoms in tetrathionate ion
(1) +6 and 0 only
(2) +6 only
(3) +5 and 0 only
(4) +5 and +6 only

49. Given below are two statements
Statement I: Oxidation state of terminal Br in Br_3O_8 is +7.
Statement II: Average oxidation number of Br in Br_3O_8 is $\frac{+16}{3}$.
In the light of above statements, choose the most appropriate answer from the options given below.
(1) Statement I is correct but statement II is incorrect
(2) Statement I is incorrect but statement II is correct
(3) Both statement I and statement II are correct
(4) Both statement I and statement II are incorrect

50. Given below are the two statements
Statement I: Species that loses electron is called reducing agent during a redox reaction.
Statement II: Species that gains electron is called oxidising agent during a redox reaction.
In the light of above statements, choose the most appropriate answer from the options given below
(1) Statement I is correct but statement II is incorrect
(2) Statement I is incorrect but statement II is correct
(3) Both statement I and statement II are correct
(4) Both statement I and statement II are incorrect

51. Consider the data given below
$Cr^{3+} + 3e^- \rightarrow Cr(s)\ E^\circ = -0.74V$
$Na^+ + e^- \rightarrow Na(s)\ E^\circ = -2.71V$
$Pb^{2+} + e^- \rightarrow Pb(s)\ E^\circ = -0.13V$
The **correct** statement is
(1) Cr^{3+} can oxidise both Na and Pb.
(2) Pb^{2+} can oxidise Na but not Cr.
(3) Na can reduce both Pb^{2+} and Cr^{3+}.
(4) Pb can reduce both Na^+ and Cr^{3+}.

52. Match reactions in List-I with their type in List-II.

List-I	List-II
(a) $KClO_3 \xrightarrow{\Delta} KCl + O_2$	(i) Non-redox
(b) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$	(ii) Non-metal displacement redox
(c) $Na + H_2O \rightarrow NaOH + H_2$	(iii) Metal displacement redox
(d) $CaCO_3 \xrightarrow{\Delta} CaO + CO_2$	(iv) Decomposition redox

Correct match is
(1) (a) – (iii), (b) – (iv), (c) – (i), (d) – (ii)
(2) (a) – (iv), (b) – (iii), (c) – (i), (d) – (ii)
(3) (a) – (iii), (b) – (iv), (c) – (ii), (d) – (i)
(4) (a) – (iv), (b) – (iii), (c) – (ii), (d) – (i)

53. Match List I with List II.

List I (Species)	List II (Oxidation number of underlined element)
a. $K\underline{Cl}O_3$	(i) +2
b. $\underline{O}F_2$	(ii) +6
c. $H_2\underline{S}O_5$	(iii) –1
d. $Na\underline{H}$	(iv) +5

Choose the correct option.
(1) a(ii), b(iii), c(iv), d(i)
(2) a(iv), b(i), c(ii), d(iii)
(3) a(i), b(ii), c(iii), d(iv)
(4) a(iv), b(iii), c(ii), d(i)

54. A hydrogen gas electrode is made by dipping platinum wire in a solution of HCl of pH = 5 and by passing hydrogen gas around the platinum wire at one atm pressure at 298 K temperature. The reduction potential of the electrode would be
(1) –0.295 V
(2) –0.148 V
(3) +0.295 V
(4) +0.148 V

55. If for the electrochemical cell
A(s) | A^{2+} (0.1 M) || B^+(0.1 M) | B(s) the emf is 5 V, then E°_{cell} of the reaction is
(1) 6.78 V
(2) 5.03 V
(3) 2.18 V
(4) 10.07 V

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56. The molar conductivity of M/10 solution of a weak monobasic acid is 20 S cm^2 mol^{-1}. If the molar conductivity at infinite dilution is 600 S cm^2 mol^{-1}, then dissociation constant of the acid is nearly

(1) 1×10^{-4} (2) 2×10^{-4}
(3) 3×10^{-5} (4) 2×10^{-5}

57. On electrolysis of an aqueous solution of $AgNO_3$ using inert electrodes (Pt), the product obtained at anode is

(1) NO_2 gas (2) N_2O gas
(3) O_2 gas (4) N_2 gas

58. Consider the following statements regarding electrochemical series

(i) A negative value of standard reduction potential of M^{n+}/M means M is a stronger reducing agent than H_2.
(ii) Fluoride ion is the weakest reducing agent.
(iii) Lithium is the strongest reducing agent.
The correct statement(s) is/are

(1) (i) only (2) (ii) only
(3) (ii) and (iii) only (4) (i), (ii) and (iii)

59. 9 g aluminium (Atomic weight 27) is deposited from aq. solution of its salt at cathode by passing certain quantity of electric charge. The volume of chlorine produced at S.T.P. by the same quantity of charge will be

(1) 22.4 L (2) 11.2 L
(3) 5.6 L (4) 44.8 L

60. Match batteries in list-I with their electrolyte in List-II.

	List-I		List-II
(a)	Dry cell	(i)	38% aq H_2SO_4
(b)	Mercury cell	(ii)	Paste of NH_4Cl and $ZnCl_2$
(c)	(H_2–O_2) Fuel cell	(iii)	Paste of KOH and ZnO
(d)	Lead storage battery	(iv)	Conc. (aq) NaOH

Correct match is

(1) (a) (ii), (b) (iii), (c)(i), (d)(iv)
(2) (a) (iii), (b) (ii), (c)(i), (d) (iv)
(3) (a)(iii), (b) (ii), (c) (iv), (d) (i)
(4) (a) (ii), (b) (iii), (c) (iv), (d) (i)

61. The limiting molar conductivity of which ion is highest in water at 298 K?

(1) Na^+ (2) K^+
(3) Mg^{2+} (4) H^+

62. The quantity of electricity required to convert 0.5 mol of MnO_4^- into Mn^{2+} is

(1) 5 × 96500 C (2) 2.5 × 96500 C
(3) 1 × 96500 C (4) 0.5 × 96500 C

63. Given : (i) $Cu^+ + e^- \rightarrow Cu$, $E^\circ = 0.52$ V
(ii) $Cu^{2+} + 2e^- \rightarrow Cu$, $E^\circ = 0.34$ V
Standard reduction potential, E° for the reaction, $Cu^{2+} + e^- \rightarrow Cu^+$ will be

(1) 0.16 V (2) 0.30 V
(3) 0.18 V (4) 0. 86 V

64. Reaction not involve in corrosion of iron is

(1) $O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$ (l)
(2) $Fe \rightarrow Fe^{2+} + 2e^-$
(3) $Fe_2O_3 + xH_2O \rightarrow Fe_2O_3 \cdot xH_2O$
(4) $4OH^- \rightarrow O_2 + 2H_2O + 4e^-$

65. The number of electrons delivered at the cathode during electrolysis by a current of 9.65 A for 100 sec is

(1) 0.01 N_A (2) 0.1 N_A
(3) 10 N_A (4) 100 N_A

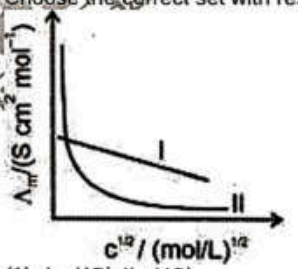
66. The specific conductance of $\frac{N}{10}$ solution of KCl at 298K is 0.003211 S cm^{-1}. If the resistance of the same solution placed in the cell is 1557.15 ohms, what is the cell constant?

(1) 6.7 (2) 5.0
(3) 5.7 (4) 8.2

67. Consider the following statements.
Statement I: Conductivity of an electrolyte decreases with dilution.
Statement II: Molar conductivity of an electrolyte increases on dilution.
In light of the above statements, identify the most appropriate answer.

(1) Statement I is correct but statement II is incorrect
(2) Statement I is incorrect but statement II is correct
(3) Both statement I and statement II are correct
(4) Both statement I and statement II are incorrect

68. Following graph has been plotted between molar conductivity (Λ_m) and $\sqrt{\text{concentration}}$ ($\sqrt{c}$) for different electrolytes.
Choose the correct set with respect to electrolyte and graph.



(1) I – KCl; II - HCl (2) I – NaCl; II – NH_4OH
(3) I – NH_4OH; II – NaCl (4) I – HCl; II – KCl

69. If conductivity of 0.2 M solution of KCl at 300 K is 0.015 S cm^{-1} then its molar conductivity (in S cm^2 mol^{-1}) is

(1) 75 (2) 50
(3) 7.5 (4) 5

70. One Faraday of electricity is passed through three electrolytic cells connected in a series containing Ag^+, Zn^{2+} and Al^{3+} ions respectively. The ratio of the moles of Ag : Zn : Al deposited at electrode is

(1) 1 : 2 : 3 (2) 3 : 2 : 1
(3) 2 : 3 : 6 (4) 6 : 3 : 2

71. 0.004 M solution of Na_2SO_4 is isotonic with a 0.0108 M solution of glucose at same temperature. The percentage dissociation of Na_2SO_4 is

(1) 75% (2) 50%
(3) 25% (4) 85%

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72. If molality of aqueous solution of glucose ($C_6H_{12}O_6$) is 1.0 m then the mass percentage of glucose in the solution will be
(1) 25.50%
(2) 10.52%
(3) 15.25%
(4) 20.25%

73. Match List-I with List-II.

	List-I		List-II Van't Hoff factor (i)
a.	K_2SO_4 with degree of dissociation 70%.	(I)	3.4
b.	$Ca_3(PO_4)_2$ with degree of dissociation 60%.	(II)	1.8
c.	$Al_2(SO_4)_3$ with degree of dissociation 75%.	(III)	2.4
d.	HCOOH with degree of dissociation 80%.	(IV)	4

Correct match is
(1) a(I), b(III), c(IV), d(II)
(2) a(I), b(III) ,c(II), d(IV)
(3) a(III), b(I), c(II), d(IV)
(4) a(III), b(I), c(IV), d(II)

74. Which of the following colligative property is widely used to determine molar mass of proteins?
(1) Relative lowering in vapour pressure
(2) Elevation in boiling point
(3) Depression in freezing point
(4) Osmotic pressure

75. An ideal solution contains 10 mol of A and 20 mol of B. The vapour pressure of the solution is (P_A° = 100 torr, P_B° = 200 torr)
(1) 133.3 torr
(2) 150 torr
(3) 166.7 torr
(4) 300 torr

76. Minimum freezing point is shown by which of the following 0.01 m aqueous solution?
(1) Urea
(2) NaCl
(3) $Al_2(SO_4)_3$
(4) $CaCl_2$

77. Boiling point of 0.1m urea(aq) solution is ($K_b(H_2O)$ = 0.52 K molal^{-1})
(1) 100°C
(2) 101°C
(3) 100.052°C
(4) 101.052°C

78. Consider the following statements.
Statement I: Solubility of gases in liquids increases with increase in temperature.
Statement II: For a solution of volatile liquids, the partial vapour pressure of each volatile component of the solution is directly proportional to its mole fraction in solution.
In the light of above statements, choose the most appropriate answer from the options given below
(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is incorrect but statement II is correct
(4) Statement I is correct but statement II is incorrect

79. The Henry's law constant (K_H) values of three gases A, B and C in water at 298 K are 1.83×10^{-5}, 40.3 and 0.413 kbar respectively. The solubility of these gases follows the order.
(1) C > B > A
(2) A > C > B
(3) B > C > A
(4) A > B > C

80. Consider the following solutions:
(a) $CH_3COCH_3 + CS_2$
(b) $C_2H_5OH + H_2O$
(c) $HNO_3 + H_2O$
(d) $C_2H_5Br + C_2H_5Cl$
The solutions which show positive deviation from Raoult's law are
(1) (a) and (b) only
(2) (b) and (c) only
(3) (a), (b) and (c) only
(4) (b), (c) and (d) only

81. Read the statements I and II carefully and select the correct option.
Statement I: Molecules of ethanoic acid dimerise in benzene due to H-bonding.
Statement II: During association of solute particles in a solution, value of colligative property decreases.
(1) Both the statements I and II are correct
(2) Both the statements I and II are incorrect
(3) Statement I is correct but Statement II is incorrect
(4) Statement I is incorrect but Statement II is correct

82. The mixture that forms maximum boiling azeotrope is:
(1) Water + Nitric acid
(2) Ethanol + Water
(3) Acetone + Carbon disulphide
(4) Heptane + Octane

83. For an ideal solution, the **correct** option is :
(1) $\Delta_{mix} S = 0$ at constant T and P
(2) $\Delta_{mix} V \neq 0$ at constant T and P
(3) $\Delta_{mix} H = 0$ at constant T and P
(4) $\Delta_{mix} G = 0$ at constant T and P

84. The osmotic pressure of equimolar solutions of glucose, KCl, $MgCl_2$ follows the order
(1) $MgCl_2$ > KCl > Glucose
(2) Glucose > KCl > $MgCl_2$
(3) KCl > $MgCl_2$ > Glucose
(4) $MgCl_2$ > Glucose > KCl

85. Boiling point of water at 760 mm Hg is 100°C. The mass of glucose added in 500 g of water such that it boils at 100.1°C is (K_b of water = 0.52 K kg mol^{-1})
(1) 17.3 g
(2) 18 g
(3) 14.2 g
(4) 21.4 g

86. A solution is prepared by adding 90 g glucose in 180 g water. The mole fraction of glucose in the solution is
(1) $\frac{1}{21}$
(2) $\frac{1}{15}$
(3) $\frac{1}{20}$
(4) $\frac{1}{16}$

87. A sample of drinking water was found to be severely contaminated with chloroform ($CHCl_3$) supposed to be a carcinogen. The level of contamination was 10 ppm (by mass). Express this in percent by mass.
(1) 10^{-3} %
(2) 10^{-4} %
(3) 8.3×10^{-6} %
(4) 8.3×10^{-5} %

88. Consider the given redox reaction in acidic medium
$MnO_4^- + C_2O_4^{2-} + H^+ \rightarrow Mn^{2+} + CO_2 + H_2O$
The number of moles of $KMnO_4$ required to oxidise 0.5 mol of oxalate ions is
(1) 0.1 mol
(2) 0.2 mol
(3) 0.5 mol
(4) 1 mol

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89. Given below are two statements, one is labelled as assertion (A) and other is labelled as reason (R).
Assertion (A): Pressure does not have any significant effect on solubility of solids in liquids.
Reason (R): Solids and liquids are highly incompressible and practically remain unaffected by changes in pressure.
In the light of above statements, choose the most appropriate answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

90. Correct expression of E°_{cell} with equilibrium constant is
(1) $E^{\circ}_{cell} = \log K_c$
(2) $E^{\circ}_{cell} = \frac{2.303\ nF}{RT} \log K_c$
(3) $E^{\circ}_{cell} = \frac{2.303\ RT}{nF} \log K_c$
(4) $E^{\circ}_{cell} = \frac{nF}{RT}$

BOTANY

91. How many of the following algae are belonging to Brown algae and green algae respectively ?
Dictyota, Volvox, Ectocarpus, Laminaria, Ulothrix, Spirogyra, Fucus, Chara, Porphyra, Polysiphonia
(1) 4, 5
(2) 4, 4
(3) 5, 4
(4) 6, 3

92. Given below are the stages in a life cycle of Moss. Arrange the following stages in the **correct** sequence.
A. Reductional division in spore mother cell
B. Protonema stage
C. Leafy stage
D. Formation of Oospore
E. Syngamy
Choose the **correct** answer from the options given below.
(1) A, B, C, D, E
(2) E, D, C, B, A
(3) B, C, D, E, A
(4) A, B, C, E, D

93. Read the following statements (a-d)
(a) These organisms can produce two kinds of spores.
(b) Sporophylls may form distinct, compact structures called strobili or cones.
(c) Fusion of male gamete with the egg present in the archegonium result in the formation of zygote.
(d) The female gametophytes are retained on the parent sporophytes permanently.
Select the features that are common in *Salvinia* and *Cycas*.
(1) Only (a) and (b)
(2) Only (a) and (c)
(3) Only (c) and (d)
(4) Only (b) and (c)

94. How many of the following statements are correct?
a. Pollen grain is reduced male gametophyte in gymnosperms.
b. Archegonia is female sporophyte in pteridophytes.
c. The male or female strobili are borne on the same tree in *Cycas*.
d. Ovules are present only in angiosperms.
e. In gymnosperms, pollens are transferred via air currents.
(1) Three
(2) Two
(3) Five
(4) Four

95. Meiosis occurs in zygote of
(1) *Marchantia*
(2) *Salvinia*
(3) *Cycas*
(4) *Spirogyra*

96. Observe the following diagram.



Select the **correct** statement with regards to the plant in the given figure.
(1) It shows the presence of cones
(2) Gametophyte is the main plant body
(3) This plant lacks vascular tissues
(4) This plant is homosporous in nature

97. Mark the **incorrect** option w.r.t. brown algae.
(1) They show great variation in size and form
(2) Plant body is differentiated into holdfast, stipe and frond
(3) They possess the photosynthetic pigments chlorophyll *a* and chlorophyll *b* but not fucoxanthin
(4) They are found primarily in marine habitats

98. In *Ulothrix*, the zygote is
(1) Formed by anisogamy
(2) Source for a hydrocolloid
(3) Subjected to complex post fertilisation development
(4) Not developed into embryo

99. Vascular archegoniates include
(1) All embryophytes
(2) All angiosperms
(3) Pteridophytes and gymnosperms
(4) Gymnosperms and angiosperm

100. An algae 'X' was reported with the following features
a. Cell wall is composed of cellulose and a hydrocolloid called carrageen.
b. Sexual reproduction is oogamous type with non motile gametes.
Algae 'X' should belong to the class
(1) Chlorophyceae
(2) Cyanophyceae
(3) Rhodophyceae
(4) Phaeophyceae

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101. Which of the following features, is **not** true for *Cycas*?
(1) Has pinnately compound leaves
(2) Its specialised roots have symbiotic association with nitrogen fixing bacteria
(3) Embryo develops inside integumented megasporangium
(4) Male and female gametophytes are produced on the same plant

102. Identify following statements as true (T) or false (F) and choose the **correct** option accordingly.
(A) In bryophytes, sporophyte derives nourishment from gametophyte
(B) Bryophytes are also known as 'Reptiles of plant-kingdom'
(C) In bryophytes sexual reproduction is isogamous and oogamous type

	(A)	(B)	(C)
(1)	T	T	T
(2)	F	T	T
(3)	T	F	F
(4)	T	T	F

(1) (1) (2) (2)
(3) (3) (4) (4)

103. Read the given Assertion (A) and Reason (R) and choose the **correct** option.
Assertion (A): A liverwort, *Marchantia* is said to be a dioecious species.
Reason (R): In *Marchantia*, sex organs antheridia (male) and archegonia (female) are produced on different thalli.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) Only (A) is true
(4) Both (A) and (R) are false

104. Select the **correct** option w.r.t. motility and origin of asexual spores of green algae
(1) Motile and endogenous
(2) Non motile and endogenous
(3) Motile and exogenous
(4) Non-motile and exogenous

105. Select the **correct** option w.r.t. *Marsilea*.
(1) Heterosporous vascular spermatophyte
(2) Heterosporous vascular cryptogam
(3) Homosporous vascular cryptogam
(4) Heterosporous non-vascular embryophyte

106. Read the assertion (A) and reason (R) given below and choose the **correct** option.
Assertion (A): In gymnosperms, seeds are not enclosed within the fruit wall.
Reason (R): Ovules become transformed into seeds as the female gametophyte is retained within the ovule on the sporophyte plant of gymnosperm.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

107. Pyrenoids are
(1) Storage bodies in most of the green algae
(2) Double membrane bound cell organelles
(3) Protein synthesising bodies in all the members of algae
(4) Complex carbohydrates in the cell wall of algae

108. Select the **odd** one w.r.t. heterospory.
(1) Important step in evolution, as it leads to seed habit
(2) Found in few pteridophytes and all gymnosperms
(3) Spores germinate to develop into dioecious sporophytic stage
(4) In plants showing heterospory, zygote development occurs within the female gametophyte

109. Read the statements given below and choose the **correct** option.
Statement I: Development of embryo occurs in all the members of Pteridophyta.
Statement II: Presence of macrophylls is a characteristic feature of *Selaginella*.
(1) Both the statement I and statement II are correct
(2) Both the statement I and statement II are incorrect
(3) Only statement I is correct
(4) Only statement II is correct

110. Identify the **mismatched** pair from the following.
(1) *Sequoia* – One of the tallest tree species
(2) *Fucus* – Rich source of iodine
(3) *Pinus* – Presence of sunken stomata
(4) *Dictyota* – Complex post-fertilization development inside the archegonium

111. Observe the plant shown in the figure given below and choose the option which is **not** true w.r.t. this plant



(1) Sporophylls form distinct compact structure called strobili
(2) The sporangia produce spores by the process of meiosis in spore mother cells
(3) Belongs to the class Pteropsida
(4) Requires water for fertilization to occur

112. Read the statements given below and choose the **correct** option.
Statement I: Bryophytes, in general, are of very little ecological importance.
Statement II: Predominant stage of the life cycle of a moss comprises of a leafy stage that develops directly from a spore.
(1) Both statements I and II are correct
(2) Both statements I and II are incorrect
(3) Only statement I is correct
(4) Only statement II is correct

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113. Select the **correctly** matched pair.
(1) Sphenopsida – *Selaginella*
(2) Pteropsida – *Adiantum*
(3) Lycopsida – *Dryopteris*
(4) Psilopsida – *Equisetum*

114. Identify the **incorrect** match.
(1) *Chlamydomonas* – Stores food as starch
(2) *Fucus* – Rarely found in fresh water
(3) *Polysiphonia* – Majorly contains fucoxanthin
(4) *Volvox* – Cellulosic cell wall

115. Select the **correct** option w.r.t. ploidy level of capsule, seta and rhizoids respectively in *Funaria*.
(1) 2n, 2n and n (2) n, n and 2n
(3) 2n, n and 2n (4) n, 2n and n

116. Natural classification system developed by Bentham and Hooker is based on
(1) Only external feature
(2) External and internal feature
(3) Only anatomy, embryology and phytochemistry
(4) External feature with phylogeny

117. Which of the following is **wrongly** matched?
(1) *Marchantia* – Free living gametophyte
(2) *Funaria* – Protonema stage
(3) *Volvox* – Colonial alga
(4) *Polysiphonia* – Isogamous reproduction
(1) (1) (2) (2)
(3) (3) (4) (4)

118. Match the following columns and select the **correct** option regarding bryophytes.

	Column I		Column II
(a)	Gametophyte	(i)	Haploid, antherozoid producing structure
(b)	Sporophyte	(ii)	Flask shaped haploid structure
(c)	Antheridium	(iii)	Produces gametes
(d)	Archegonium	(iv)	Non photosynthetic diploid body

(1) a(iii), b(iv), c(ii), d(i) (2) a(ii), b(i), c(iii), d(iv)
(3) a(iii), b(iv), c(i), d(ii) (4) a(iv), b(iii), c(i), d(ii)

119. Unlike pteridophytes, gymnosperms do **not** have
(1) Independent free living male and female gametophytes
(2) Differentiated main plant body
(3) Diploid sporophytic phase in the life cycle
(4) Vascular tissues

120. The ovules of gymnosperms remain exposed
(1) Before formation of ovary and fruits
(2) After fertilization only
(3) Both before and after fertilization
(4) After triploid endosperm formation

121. In gymnosperms the male gametophyte is highly reduced, known as
(1) Endosperm (2) Embryo sac
(3) Nucellus (4) Pollen grain

122. 2-8 equal and apical flagella in zoospores is found in the members of
(1) Cyanobacteria (2) Green algae
(3) Red algae (4) Brown algae

123. Mark the **odd** one w.r.t. ploidy level in mosses.
(1) Zygote (2) Capsule
(3) Protonema (4) Spore mother cell

124. The phenomenon of pollination is found in which of the following plant groups?
(1) Algae (2) Gymnosperms
(3) Bryophytes (4) Pteridophytes

125. Leaves of conifers do **not** have
(1) Needle like shape
(2) Very thin cuticle
(3) Sunken stomata
(4) Adaptation to withstand extreme temperature

126. Green algae lack
(1) Cell wall
(2) Chlorophyll molecules
(3) Motile spores
(4) Coating of algin in cell wall

127. Gemmae are
(1) Non green, multicellular, asexual buds
(2) Green, multicellular, asexual buds
(3) Non green, unicellular, asexual buds
(4) Non green, multicellular, sexual buds

128. Coralloid roots are found in
(1) *Pinus* (2) *Cycas*
(3) *Pteris* (4) *Adiantum*

129. *Pinus* is ___(a)___ plant and has ___(b)___ stem
Select the correct option to fill in the blanks (a) and (b).

	(a)	(b)
(1)	Monoecious	Branched
(2)	Dioecious	Unbranched
(3)	Monoecious	Unbranched
(4)	Dioecious	Branched

(1) (1) (2) (2)
(3) (3) (4) (4)

130. Agar is obtained from
(1) *Spirogyra* (2) *Laminaria*
(3) *Dictyota* (4) *Gelidium* and *Gracilaria*

131. The food is stored as floridean starch in
(1) *Ectocarpus* (2) *Polysiphonia*
(3) *Chara* (4) *Spirulina*

132. The female gametophyte in _____ is retained on the parent sporophyte for variable periods.
Choose the **correct** option to fill in the blank.
(1) *Salvinia* (2) *Dryopteris*
(3) *Lycopodium* (4) *Adiantum*

133. Each character is given equal importance and at the same time, hundreds of characters can be considered in
(1) Phenetics (2) Chemotaxonomy
(3) Cytotaxonomy (4) Karyotaxonomy

134. Select the **incorrect** statement w.r.t. artificial system of classification.
(1) It gave equal weightage to vegetative and sexual characteristics
(2) It is based on a few morphological characters
(3) It is an advanced system of classification
(4) It separated the closely related species

135. Linnaeus artificial classification system of angiosperms was based on
(1) Stomata in leaves
(2) Internal structure of stem
(3) Modification of different parts of plants
(4) Androecium structure

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136. Which of the following statements is incorrect?
(1) Hormones are required in small quantities
(2) Hormones are released into the blood
(3) Hormones are highly specific in their function
(4) Hormones provide energy or building materials

137. All of the following are differences between neural system and endocrine system, **except**
(1) Former provides point to point coordination
(2) Only former provides homeostasis
(3) Former provides short-lived coordination
(4) Former provides rapid coordination

138. Select the secretion which is poured into the interstitial fluid surrounding the secretory cells rather than into ducts.
(1) Milk (2) Sweat
(3) Melatonin (4) Sebum

139. Pineal gland is located on the
(1) Ventral side of forebrain (2) Lateral side of forebrain
(3) Dorsal side of forebrain (4) Ventral side of midbrain

140. Largest endocrine gland in the human body is
(1) Thyroid gland (2) Adrenal gland
(3) Thymus (4) Pituitary gland

141. In males, _____ stimulates the synthesis and secretion of androgens. In females, _____ stimulates the growth and development of ovarian follicles.
Choose the option that **correctly** fill the blanks respectively.
(1) LH, FSH
(2) GnRH, prolactin
(3) Testosterone, progesterone
(4) ADH, relaxin

142. Glucocorticoids promote all, **except**
(1) Gluconeogenesis
(2) Lipolysis
(3) Proteolysis
(4) Storage of glucose in liver

143. Secretions of endocrine glands named hormones enter blood capillaries and circulate through out body. Choose the pair of purely endocrine glands.
(1) Adrenal gland and testes
(2) Pineal gland and adrenal gland
(3) Pancreas and liver
(4) Thymus and ovary

144. Oxytocin and vasopressin are transported from hypothalamus to
(1) Adenohypophysis through axon
(2) Neurohypophysis through axon
(3) Adenohypophysis through blood
(4) Neurohypophysis through blood

145. Body under the influence of catecholamines will not show
(1) Increase in blood glucose level
(2) Increase in heart beat
(3) Increase in rate of respiration
(4) Increase in protein anabolism

146. Thymus gland is a lobular structure located between lungs behind _____ on the _____ side of aorta.
Choose the **correct** option which fills the blanks respectively.
(1) Clavicle, dorsal (2) Clavicle, lateral
(3) Sternum, ventral (4) Sternum, dorsal

147. Which of the following hormones is **not** synthesized by pituitary gland?
(1) Gonadotropins (2) Somatostatin
(3) Somatotropin (4) Prolactin

148. All of the following are the symptoms of hyperthyroidism, **except**
(1) Excessive weight loss
(2) Low intelligent quotient
(3) Increased basal metabolic rate
(4) Enlargement of thyroid gland

149. Excess of GH during adulthood results in
(1) Acromegaly (2) Acromicria
(3) Atelectasis (4) Ateleosis

150. Mineralocorticoid secreted by adrenal cortex among the following is
(1) Cortisol (2) Aldosterone
(3) Vasopressin (4) Adrenaline

151. Which of the following hormones released from adrenal gland has membrane-bound receptors on their respective target cells?
(1) Cortisol (2) Aldosterone
(3) Epinephrine (4) Androgenic steroids

152. Which of the following hormones is known as emergency hormone?
(1) Adrenaline
(2) Angiotensinogen
(3) Aldosterone
(4) Adrenocorticotropin hormone

153. The flap of connective tissue that connects the two lobes of thyroid gland is called
(1) Infundibulum (2) Islet
(3) Isthmus (4) Intercalated disc

154. In which disease, underproduction of hormone alters carbohydrate metabolism causing acute weakness and fatigue and led to bronze like pigmentation of skin?
(1) Diabetes insipidus (2) Myasthenia gravis
(3) Addison's disease (4) Graves' disease

155. Cell mediated immunity is provided by T-lymphocytes. Which hormone is primarily responsible for the differentiation and functioning of T-lymphocytes?
(1) Thyroxine (2) Thymosin
(3) Vasopressin (4) Thyrocalcitonin

156. Which of the following is **not** a function of progesterone?
(1) It maintains a secretory corpus luteum
(2) Formation of alveoli in mammary glands
(3) Supports pregnancy and thus prevents abortion
(4) Stimulates secretion of milk

157. The hormone antagonistic to parathormone is secreted by
(1) Thyroid gland (2) Thymus
(3) Pineal gland (4) Parathyroid gland

158. Androgenic steroids are secreted by Leydig cells in males but small amounts of them are also secreted by
(1) Endocrine pancreas (2) Adrenal cortex
(3) Adrenal medulla (4) Thyroid gland

159. All of the following hormones play distinct and significant roles in regulation of the body temperature, **except**
(1) Melatonin (2) Thyroxine
(3) Growth hormone (4) Thyrocalcitonin

160. The hormone which stimulates secretion of water and bicarbonate ions from exocrine pancreas in humans is
(1) Gastrin (2) Secretin
(3) Cholecystokinin (4) GIP

161. Complete the analogy by choosing the correct option.
Steroid hormone : Progesterone : : Amino-acid derivative : _____
(1) Epinephrine (2) Growth hormone
(3) TSH (4) Erythropoietin

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162. Ovarian growth is physiological response of FSH which acts on receptors to generate a 2nd messenger named
(1) cGMP (2) IP_3
(3) DAG (4) cAMP

163. ANF is a peptide hormone secreted from
(1) Atrial wall (2) Ventricular wall
(3) JG cells (4) Cells in blood vessels

164. Choose the **odd** one w.r.t. hormones of gastrointestinal tract.
(1) Gastric inhibitory peptide (2) Cholecystokinin
(3) Secretin (4) Trypsinogen

165. Which of the following hormones acts on both pancreas and gall bladder and stimulates the secretion of enzymes in pancreatic juice from pancreas and the release of bile juice from gall bladder?
(1) Gastrin (2) Secretin
(3) Cholecystokinin (4) Gastric inhibitory peptide

166. All of the following hormones require second messengers for their action **except**
(1) Hormones of adrenal cortex
(2) All polypeptide and protein hormones
(3) Epinephrine
(4) Hypothalamic hormones

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167. Hormone GIP acts on
(1) Liver (2) Pancreas
(3) Intestinal glands (4) Gastric glands

168. Several groups of neurosecretory cells which produce hormones to regulate the synthesis and secretion of hormones of adenohypophysis, are called
(1) Nerve plexuses (2) Neuroglia
(3) Nuclei (4) Ganglia

169. All of the following hormones influence the cellular activities by interaction of hormone-receptor complex with the genome, **except**
(1) Glucocorticoids (2) Testosterone
(3) Estradiol (4) Oxytocin

170. The effect of GnRH produced by the hypothalamus directly stimulates the
(1) Synthesis and secretion of gonadotrophins
(2) Secretion of milk in mammary glands
(3) Synthesis of carbohydrates
(4) Fetal ejection reflex

171. A person passes a lot of dilute urine and drinks a lot of water but does not have glycosuria. He is most likely suffering from
(1) Diabetes mellitus (2) Diabetes insipidus
(3) Cretinism (4) Gigantism

172. In humans, which of the following is considered as master clock of the body?
(1) Hypothalamus (2) Thymus
(3) Adrenal gland (4) Hypophysis

173. Parathormone deficiency produces tetany due to
(1) Enhanced Na^+ in blood
(2) Enhanced glucose in blood
(3) Enhanced Ca^+ in blood
(4) Decreased level of Ca^+ in blood

174. Which of the following is **not** a function of LH?
(1) Induces ovulation of Graafian follicle
(2) Maintains corpus luteum
(3) Induces Leydig cells to secrete androgens
(4) Acts on cells of Sertoli for spermatogenesis

175. Pituitary gland is **not** responsible for synthesis of
(1) LH (2) FSH
(3) Oxytocin (4) GH

176. In mammals, the female secondary sexual characteristics are developed by
(1) Relaxin (2) Estrogens
(3) Progesterone (4) Gonadotropins

177. How many of the hormones given in the box below is/are considered as trophic hormones?

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Prolactin, Oxytocin, Vasopressin, Luteinizing hormone, Follicle stimulating hormone

(1) Four (2) Three
(3) Two (4) One

178. **Assertion (A):** Prolactin regulates the growth of mammary glands and formation of milk in them.
Reason (R): Prolactin is produced by pars nervosa and acts on smooth muscles of the mammary glands.
In the light of above statements, choose the **correct** answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

179. Symptoms such as stunted growth, mental retardation, a low metabolic rate and deaf-mutism in a child indicate that the pregnant mother suffered from hyposecretion of hormone produced from
(1) Parathyroid gland (2) Thyroid gland
(3) Pancreas (4) Liver

180. Choose the hormone which is secreted by the gland that is not located in the cranial cavity but causes gluconeogenesis.
(1) Insulin (2) Epinephrine
(3) Melatonin (4) ACTH

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